

Sales and Purchases Data Analysis Report (Cottonil)

An academic report presenting a Power BI-based analysis of Cottonil's transactional data (sales, purchases, products, suppliers, and time series). This deck summarizes methods, dashboards, visualizations, key findings, and actionable insights for instructors and graduate students in data science and public health.



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About Cottonil

Cottonil is one of the largest and most prominent Egyptian companies specializing in the manufacturing of **underwear and cotton products**. The company was founded in **1999** by businessman **Basel Somakia**, and quickly established itself as a leading brand in the Egyptian market.



Overview



Sales



Purchases

50%
EGYPTIAN
MARKET SHARE



Data Overview

The dataset used in this project was collected from a real Cottonil retail store. The data was originally in HTML format and was converted into Excel format before being cleaned and processed using Power BI.

The dataset includes information about sales, purchases, products, suppliers, and time-based transactions. This real-world dataset provides valuable insights into business performance and operational trends.

Analysis Process

Data Cleaning & Transformation

The dataset required extensive preprocessing before analysis. The raw data was originally collected in HTML format and was converted into Excel to enable structured processing.

Several data cleaning steps were performed, including:

- Removing duplicate records
- Standardizing product and supplier names
- Handling missing values
- Correcting inconsistent formats (especially dates and text fields)

These steps were essential to ensure data quality, accuracy, and consistency before performing the analysis in Power BI.



Data Preparation & Analysis

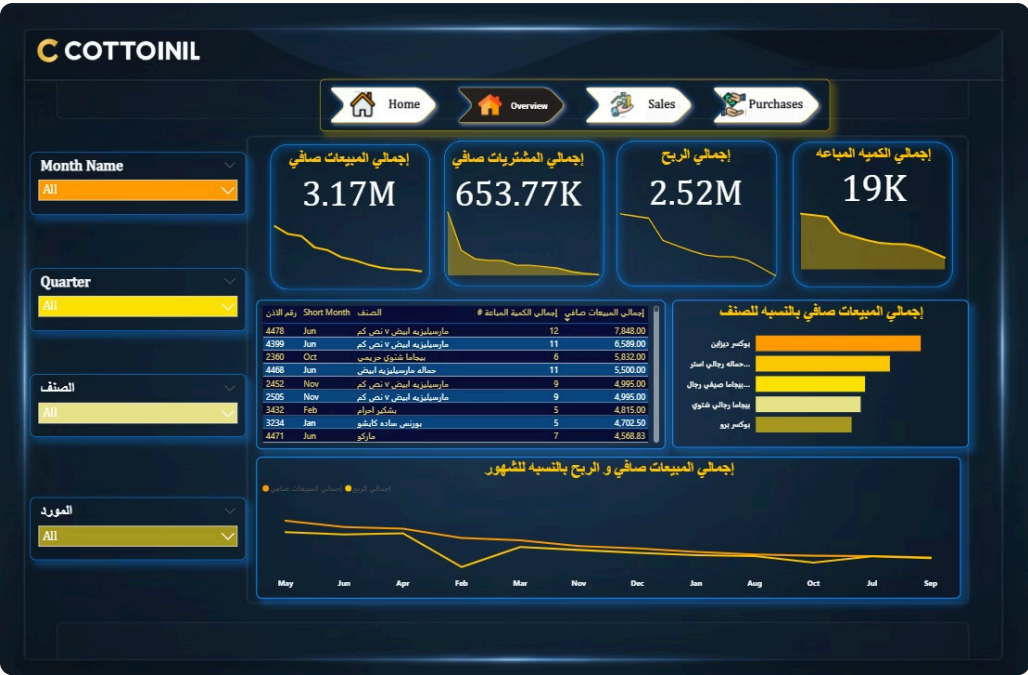
◆ Data Preparation

After cleaning, the dataset was prepared by creating calculated measures such as total sales, total purchases, profit, and average order price. A date table was created, and relationships between tables were defined in Power BI.

◆ Data Analysis

The prepared data was analyzed using Power BI through multiple dashboard pages (Overview, Sales, and Purchases). Various visualizations such as KPIs, line charts, and bar charts were used to explore trends and answer key business questions.

Dashboard Pages (Placeholders)



Overview Dashboard

Snapshot of top-level KPIs (total sales, total purchases, profit, avg price) and summary time trend. Entry point for deeper investigation with slicers for month, category, and supplier.



Sales Dashboard

Detailed views of sales by category, product, and time. Includes KPIs, line charts for trends, and bar charts for top-selling categories and products.



Purchases Dashboard

Purchases-focused page showing purchase volume, supplier contributions, and purchase trends over time to assess cost drivers and supplier impact.

Question 1 — What is total sales vs total purchases?

Insight: Compare aggregated sales and purchases over the analysis period. Compute absolute difference and margin percentage: Profit = Sales – Purchases. This KPI reveals gross profitability and capital flow dynamics.





Question 2 — How does performance change over time?

Insight: Time-series line charts identify seasonal patterns, growth or decline phases, and anomalous months. Use rolling averages to smooth noise and month-over-month percentage change to quantify trends.

Question 3 — Which product categories perform the best?

Insight: Category-level bar charts rank performance by revenue, quantity, and profit contribution. Identifies dominant categories and long-tail performers for assortment and inventory decisions.

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Question 4 — What is the average order price?

Insight: Average Order Price = Total Sales / Number of Orders.

Complement KPI with a histogram to examine distribution (skew, outliers) and median price to mitigate influence of extreme values.



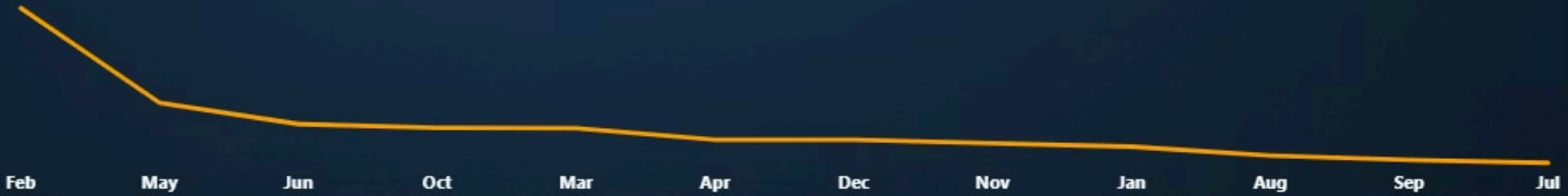
Question 5 Which product categories have the highest purchases?



Supplier Impact

Insight: Analyze supplier share of purchases, unit prices by supplier, and lead-time variability. Identify concentration risk and opportunities for negotiation.

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Question 6 What are the trends in purchases over time?

Purchase Trends

Insight: Time-series of purchases (monthly) reveals procurement cycles, spikes related to promotions or restocks, and long-term procurement cost trends.

The analysis uses multiple types of visualizations to present insights clearly:

- KPI Cards: Used to display key metrics such as total sales, total purchases, profit, and average order price.
- Line Charts: Used to analyze trends over time for sales and purchases.
- Bar Charts: Used to compare performance across product categories.
- Tables: Used to display detailed transactional data for validation and deeper analysis.

These visualizations help simplify complex data and support data-driven decision making.



Figure: Sample visualizations used in the analysis (KPI cards, bar chart, and line chart).

Visualizations, Key Findings & Dashboard Narrative



KPI Cards

Concise numeric summaries for Total Sales, Total Purchases, Profit, Avg Order Price and Quantity.



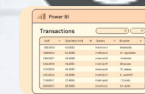
Line Charts

Time trends showing seasonality, growth, and anomalies for sales and purchases.



Bar Charts

Category and product comparisons for revenue, quantity, and profit ranking.



Tables

Drillable transaction detail for validation, audit, and cross-checking analytic aggregates.



Key Findings

- Sales exceed purchases during the analysis period, producing positive gross profit overall.
- A small number of product categories account for a large share of sales (category concentration).
- Clear temporal trends and seasonality are present; targeted months show pronounced peaks.
- Supplier composition materially affects purchase volumes and average purchase price—potential negotiation targets identified.

Dashboard Narrative & Interaction

The dashboard is designed as a multi-page storytelling tool. It starts with an overview of key performance indicators such as total sales, purchases, and profit.

It then moves to a detailed analysis of sales performance across categories and time. Finally, it explores purchase behavior and trends to provide a complete view of business operations.

Users can interact with the dashboard using filters such as month, product category, and supplier, allowing dynamic exploration and deeper insights.